

Chapter 6, Problem 7P

Problem

At $t = 0$, the voltage across a 25-mF capacitor is 10 V. Calculate the voltage across the capacitor for $t > 0$ when current $5t$ mA flows through it.

Step-by-step solution

Step 1 of 1

Given capacitor is $C = 25 \text{ mF}$ voltage $v = 10 \text{ V}$ at $t = 0$

When $i(t) = 5t \text{ mA}$ for $t > 0$

Voltage across capacitor is $v(t) = \frac{1}{C} \int i(t) dt + v(t_0)$

$$v(t) = \frac{1}{C} \int_{t_0}^t i(t) dt + v(t_0)$$

$$\begin{aligned} v(t) &= \frac{1}{25 \times 10^{-3}} \int_0^t 5t \times 10^{-3} + v(0) \\ &= \frac{1}{25 \times 10^{-3}} \left[\frac{5t^2}{2} \times 10^{-3} \right]_0^t + 10 \\ &= \left(\frac{t^2}{10} + 10 \right) \end{aligned}$$

$$v(t) = (0.1t^2 + 10)$$

Hence the voltage $v(t)$ across the capacitor is $\boxed{(0.1t^2 + 10) \text{ V}}$